14 REST API Development with Laravel

In this chapter, you will learn how to build RESTful APIs using Laravel 12. You will explore the core principles of REST, how Laravel facilitates API development, and how to build a simple yet complete API for a product catalog. The chapter also covers input validation, JSON responses, authentication using Laravel Sanctum, pagination, and testing. By the end of this chapter, you will have a solid foundation for building professional APIs using Laravel.

14.1 Introduction to REST APIs

REST (Representational State Transfer) is an architectural style that defines a set of constraints for building web services. RESTful APIs use HTTP methods such as GET, POST, PUT, and DELETE to perform operations on resources. Each resource is identified by a URL and typically represented in JSON format.

Compared to older protocols like SOAP, REST is lightweight, stateless, and easier to implement. In Laravel, RESTful APIs can be created quickly using built-in routing and controller features. A well-designed RESTful API improves client-server communication, especially in mobile and single-page applications.

14.2 Setting Up the Laravel API Project

To get started, you can either use an existing Laravel project or create a new one using the command:

```
laravel new laravel-api-demo
```

Once the project is set up, update the `.env` file with your database configuration. Make sure the database is created and accessible. Laravel uses `routes/api.php` to define routes that are specifically intended for API requests. These routes are automatically prefixed with `/api` and are stateless by default.

14.3 Defining API Routes

Laravel provides a convenient way to define RESTful routes using the `Route::apiResource` method. This automatically generates routes for the standard CRUD operations. Here is an example of defining API routes for a `ProductController`:

```
Route::apiResource('products', ProductController::class);
```